



ASSET FLIP

A 2D Platformer in Two Acts

Patrick Anderson & Alan Ray

Overview

Background

A game built entirely in **Godot 4** using **GDScript**.

The project's theme emerged organically from free assets discovered during development - one partner found Mario-style assets, the other found Hollow Knight-inspired ones.

The result is ***Asset Flip***: a game taking inspiration from multiple decades of 2D platformers.

Objectives

Demonstrate understanding of:

- Physics simulation & movement
- Finite state machines
- Collision & hit detection
- Scene/node architecture
- Signal architecture
- Iterative, sprint-based development

User Story

“

A player wants to enjoy a fast-paced 2D platformer where they can navigate challenging levels and defeat enemies in satisfying combat - experiencing a dramatic tonal shift from a bright, cheerful world into a dark, oppressive one.

”

Features

Essential (all delivered)

- Responsive movement & jumping
- Gravity, floor collision
- Double jump & dash
- Directional melee attack system
- Player FSM
- Enemy FSM
- Enemy health & death
- Room-to-room transitions
- Player health UI
- 2 distinct areas
- Menu screen
- VFX
- Audio
- Win / lose conditions

Desirable (all but last delivered)

- Boss fight with unique behavior
- Parallax scrolling backgrounds
- 3-12 enemy types
- Tutorial prompts
- Screen shake & knockback
- Pause screen
- Boss health bar

Stretch Goals (delivered first two)

- Second unique boss fight
- Movement polish (hang time at apex, coyote time)
- Controller support
- Ranged player attacks
- Complex enemy AI platforming
- Interactive items in environment
- Settings menu
- Save / load

System Architecture

7 Global Autoload Singletons

GameFlow	Scene transitions, fade overlay, win/death routing
AudioManager	Music, ambient loops, pooled SFX playback
VFX	Trauma-based screen shake
HealthUI	Pip health display; reconnects each scene load
PauseMenu	Escape key
ScoreManager	Score tracking, score-changed signal
UISound	Auto-wires hover/click sounds to every Button

Active Scene (rebuilt each transition)

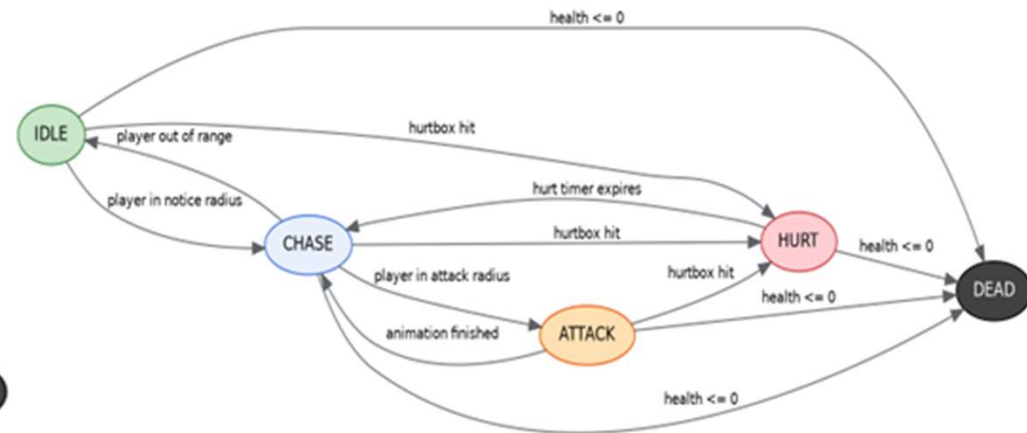
- Levels
- Menu screen
- Death/win screens
- Player
- Spawn points
- Transition areas
- Enemy instances

Finite State Machines

Player FSM (6 states)

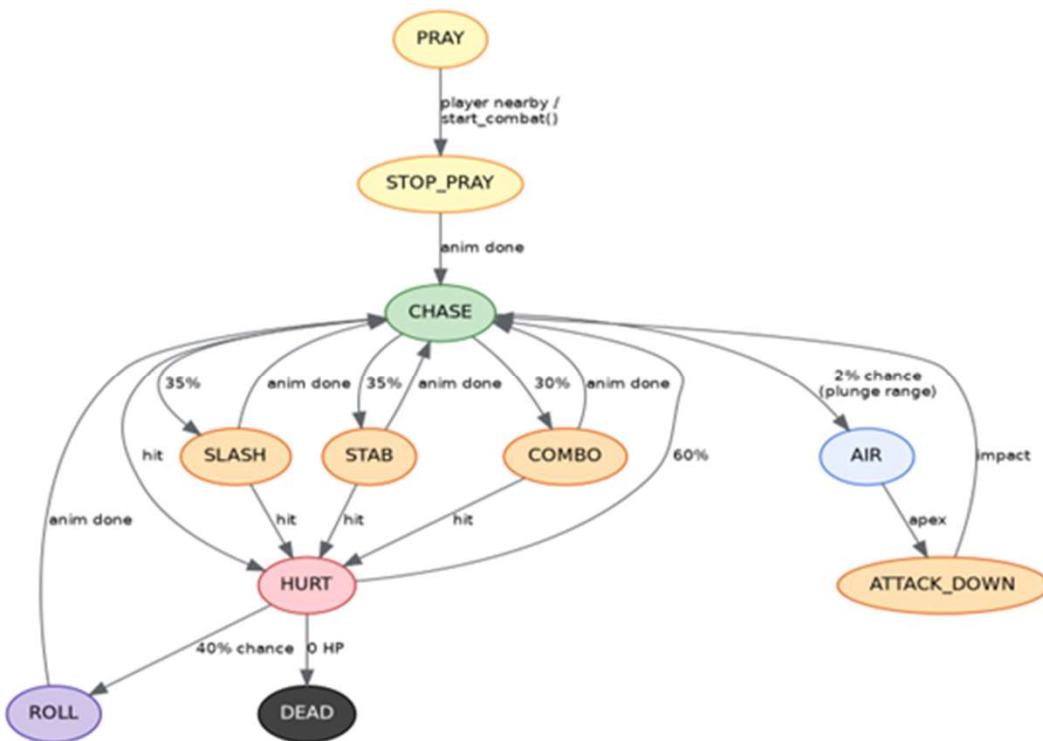


Generic Enemy FSM (5 states)

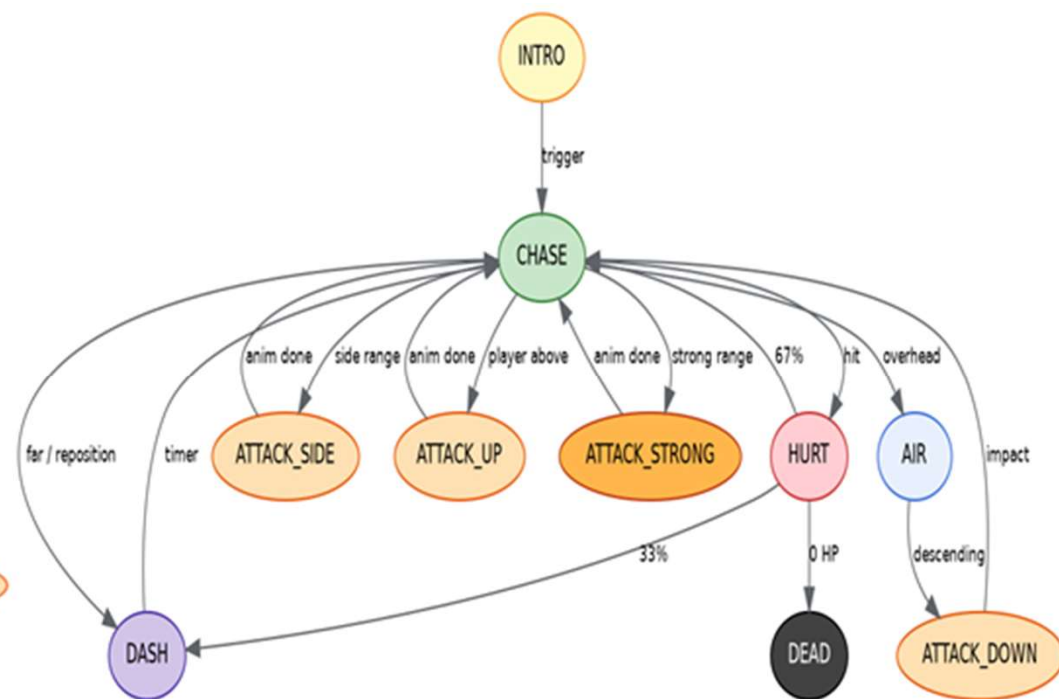


Boss Finite State Machines

Knight Boss (11 States)



Witch Boss (10 States)



The background is a solid dark blue. On the left, there is a large, semi-transparent purple circle. On the right, there is a large, semi-transparent teal circle. A thin, solid purple horizontal line is positioned below the word 'DEMO'.

DEMO

TIMELINE

Sprint Plan — 11 Weeks

Sprints 1–4

Feb 5 – Feb 26

Movement · Combat · Animations · Player & Enemy FSMs

Playable prototype by Sprint 4

Sprints 5–7

Mar 5 – Mar 19

First boss · Asset theming per area · BaseEnemy refactor · Level transitions · Parallax

Sprints 8–9

Mar 26 – Apr 2

Win/lose flows · Menus & death screens · VFX pass · Tutorial level · Bug fixes

Sprint 10

Apr 9

Finalize 8 enemies · Level layouts & lighting · Full audio bank · Asset cleanup

Sprint 11

Apr 16

Knight boss · Interface sounds · Dash VFX · Footsteps · Double-jump animation

Every week: testing, tuning, bug-fixing, level creation.

Capstone Progress

Capstone Milestone Tracking Chart

